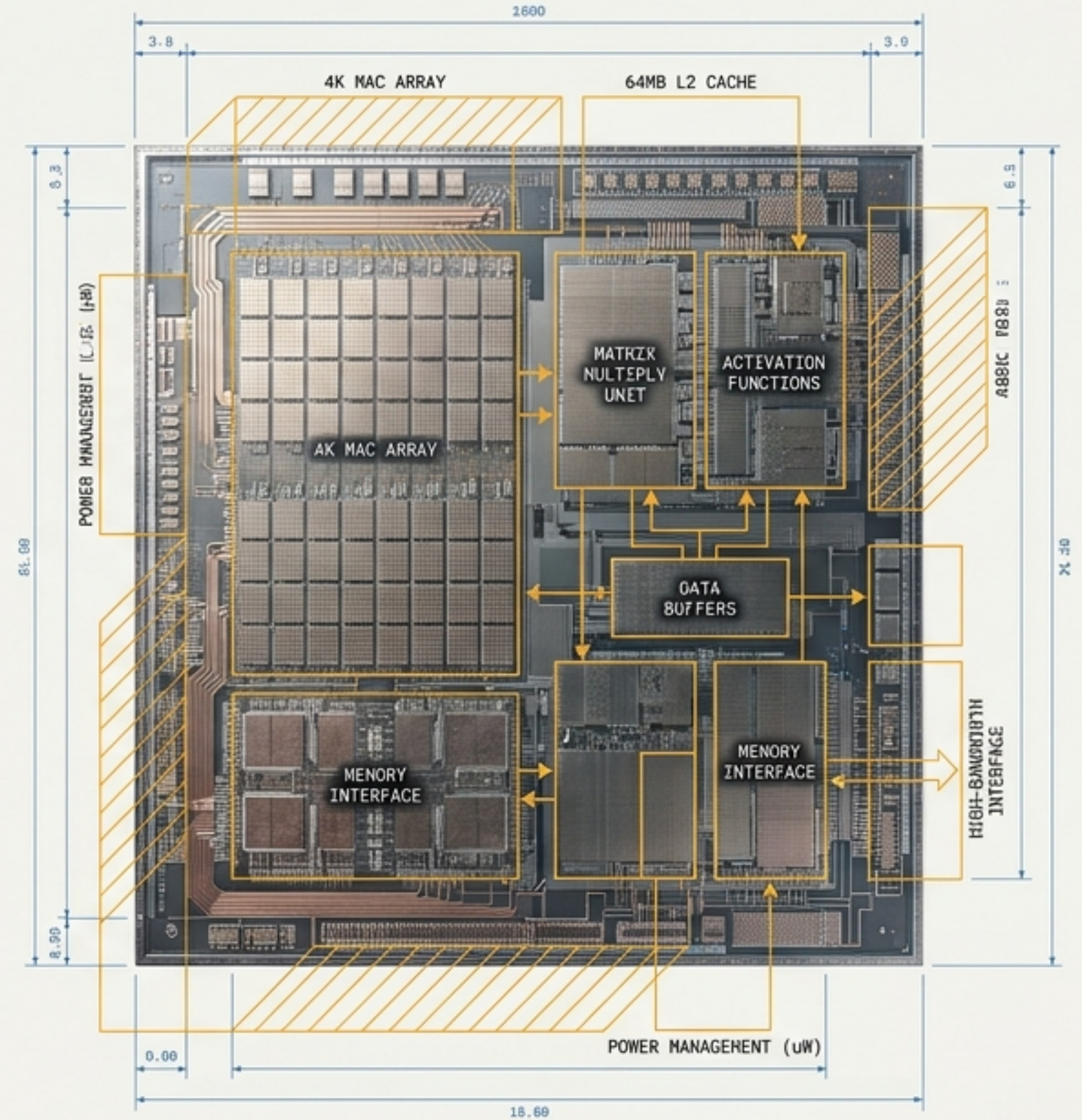
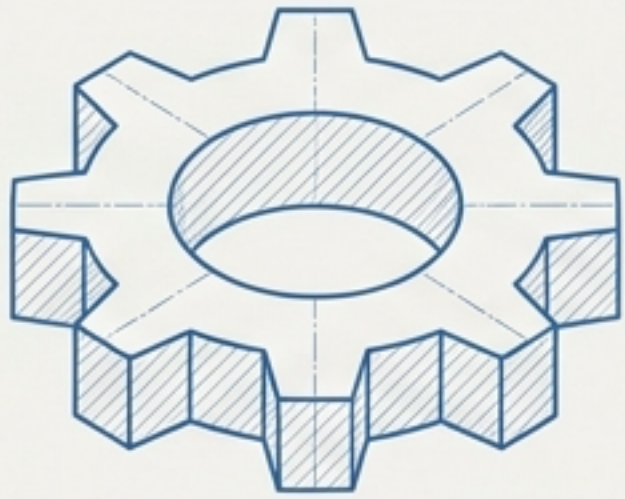


NPU: Neural Processing Unit

The specialized silicon powering the on-device AI revolution.

A dedicated hardware accelerator purpose-built to execute the mathematical workloads of artificial neural networks—trading general-purpose flexibility for unmatched power efficiency and speed.

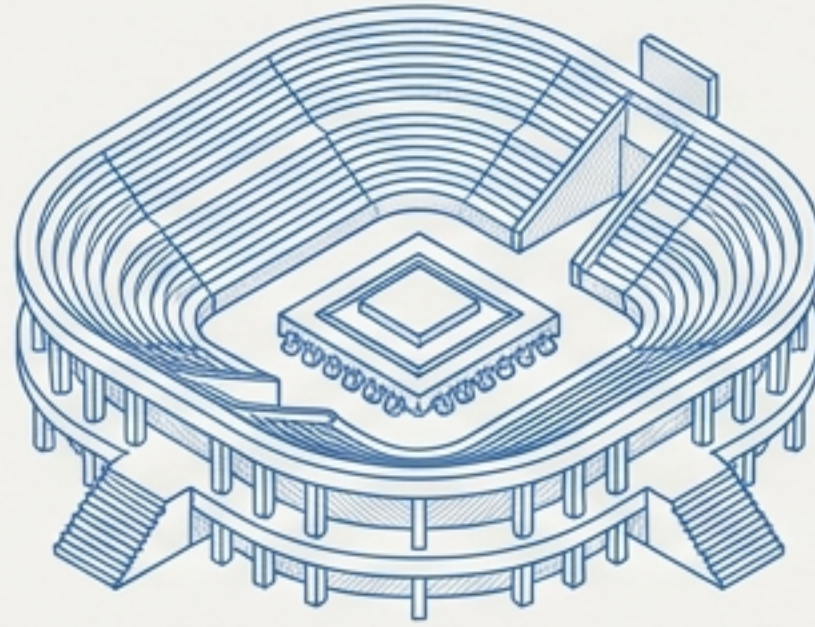




CPU (Central Processing Unit)

The Generalist

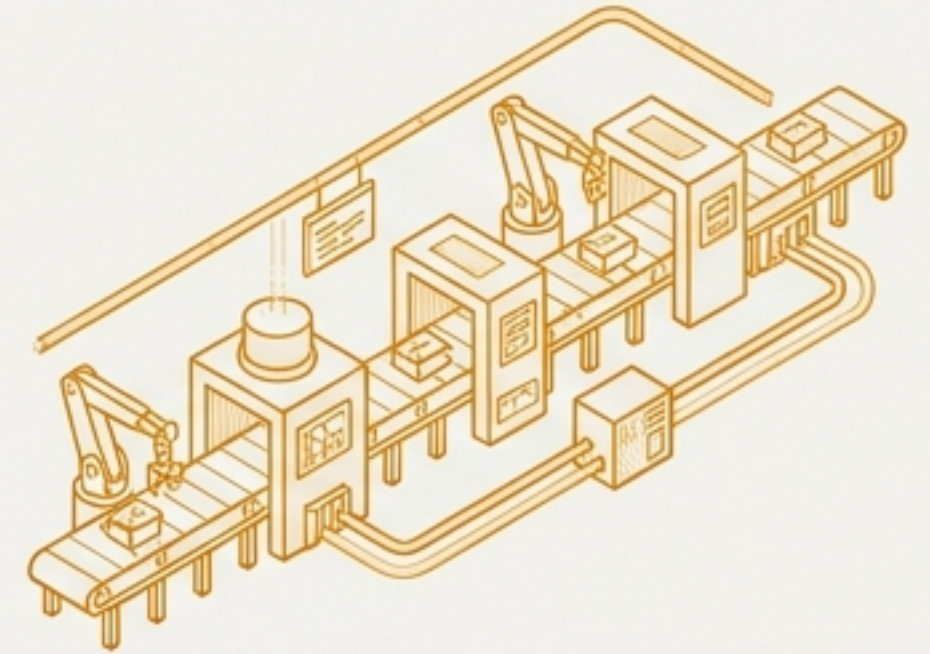
Designed for sequential control logic and versatility. High flexibility, but low efficiency for parallel AI math.



GPU (Graphics Processing Unit)

The Parallel Powerhouse

Designed for high-throughput graphics and AI model training. Massive parallelism, but massive power consumption (300W–700W in data centers).



NPU (Neural Processing Unit)

The AI Specialist

Designed exclusively for AI inference. Hyper-optimized for low-precision tensor math (INT8) and massive data reuse. Operates at a fraction of the power cost of a GPU.

1. Input

The gateway for streaming data (weights and activations) into the computation arrays.

2. Memory

The integrated on-chip storage hierarchy designed to prevent the "memory wall."

3. MAC (Multiply-Accumulate)

The core systolic array performing billions of calculations per cycle.

4. VPU (Vector Processing Unit)

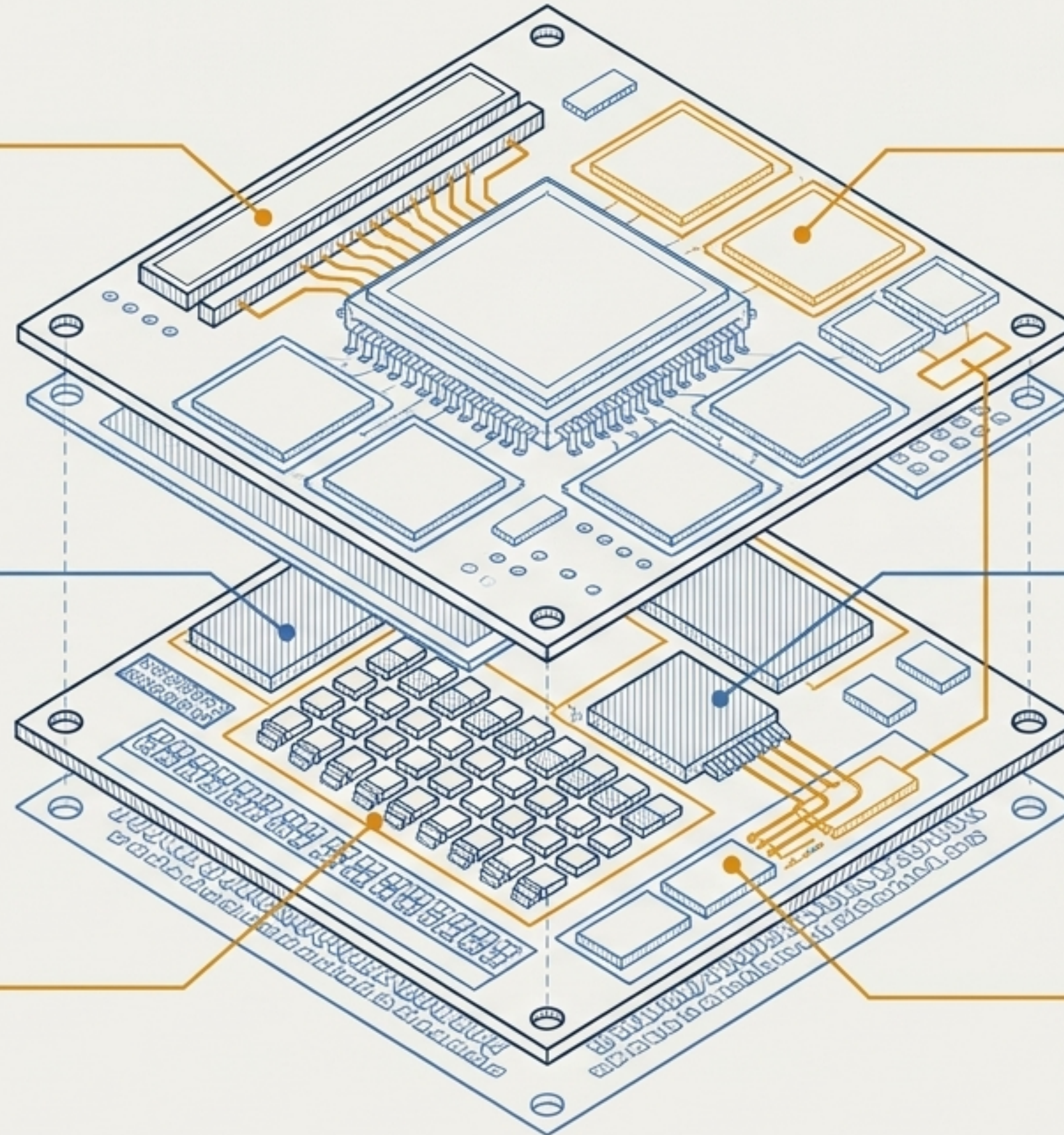
The specialized engine for non-linear mathematical functions.

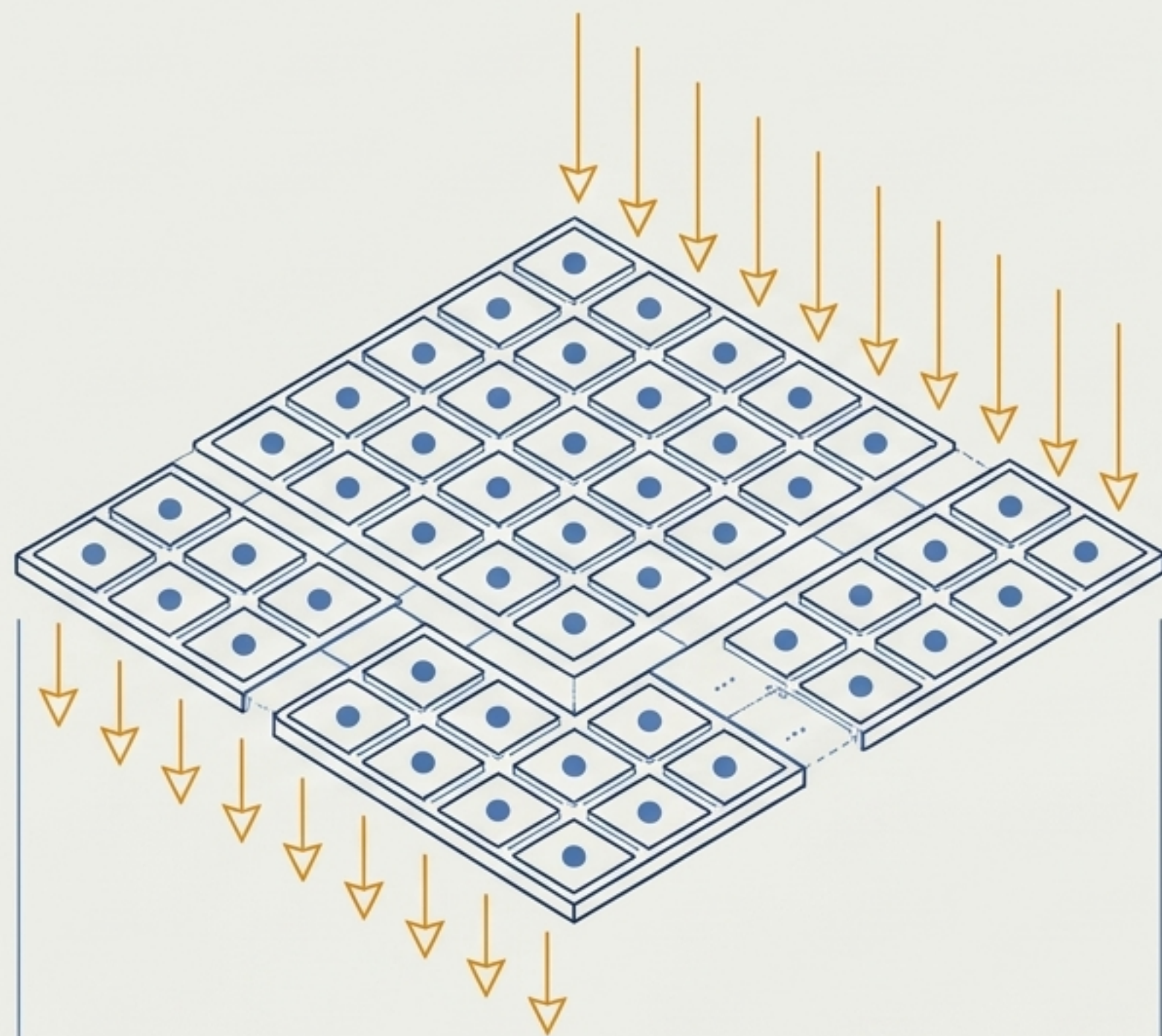
5. SRAM (Static Random Access Memory)

The ultra-fast local storage keeping data nanoseconds away from compute.

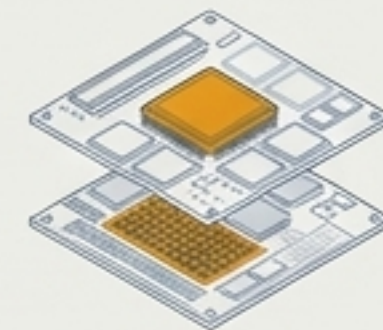
6. DAM / DMA (Data Access Management)

The intelligent controller orchestrating the flow of data across the chip.





$[\text{Weight} \times \text{Input}] + \text{Accumulator}$



MAC (Multiply-Accumulate): The Heartbeat of AI

The Concept

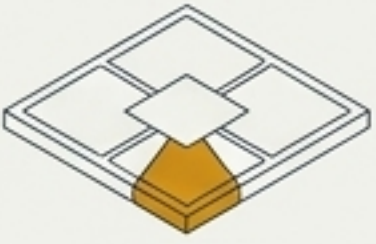
Over 90% of a neural network's workload is a single mathematical operation repeated billions of times: multiplying a weight by an input, and adding it to a running total.

The Architecture

MAC units are arranged in massive 2D grids. Data pulses through them like blood through tissue—maximizing data reuse.

The Result

An NPU executes thousands of MAC operations per clock cycle, achieving extreme TOPS (Trillions of Operations Per Second).



VPU (Vector Processing Unit): Beyond Simple Math

The Concept

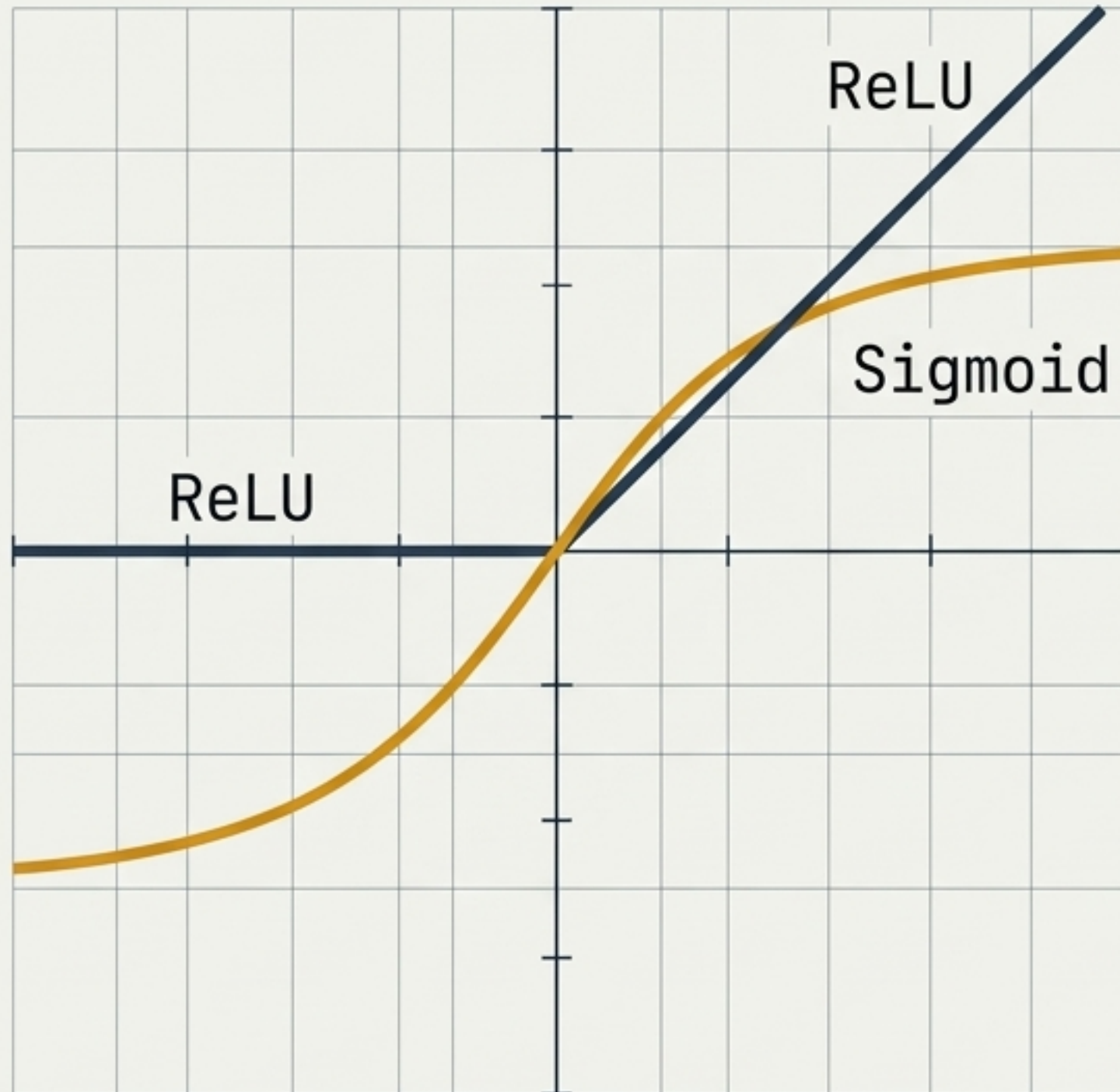
While MAC arrays excel at basic multiplication and addition, neural networks also require complex, non-linear activation functions to make intelligent decisions.

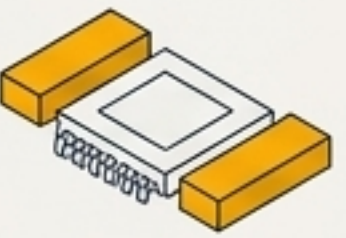
The Role

The VPU acts as a specialized co-processor within the NPU. It handles critical vector operations like Softmax, ReLU (Rectified Linear Unit), and pooling.

The Synergy

By offloading these complex functions to the VPU, the MAC arrays are freed up to continue their high-speed matrix multiplications without bottlenecking.





SRAM (Static Random Access Memory): Beating the Memory Wall

The Problem

Reading one value from off-chip RAM costs **~200 times** the energy of the MAC calculation itself and takes **100 times longer**.

The Solution

NPU's hoard data in **high-capacity, on-chip SRAM**. This keeps the AI model's **weights and activations** physically adjacent to the compute grid.

The Benefit

By minimizing data movement, SRAM **drastically cuts energy consumption and latency**, making on-device, battery-powered AI inference possible.

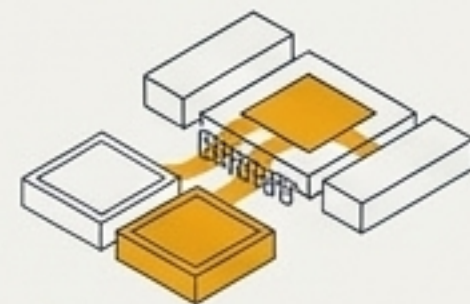
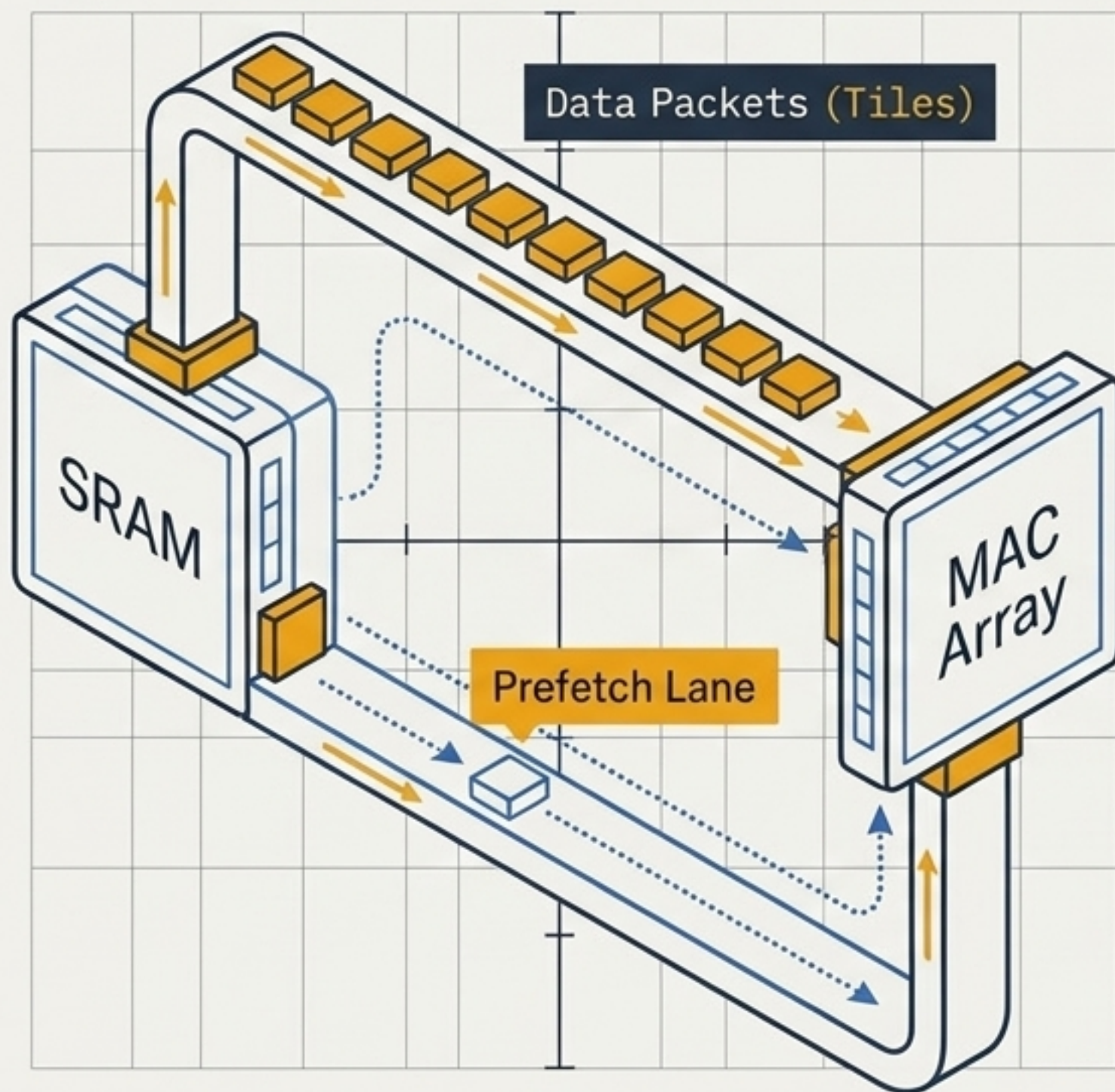
Diagnostic Scale

Off-Chip RAM

100x Longer / ~200x Energy Cost

On-Chip SRAM

Nanosecond Distance



DAM (Data Access Management / DMA)

The Concept

An NPU is only as fast as its ability to feed data to its compute engines. If the massive MAC array is starved of data, performance plummets.

The Role

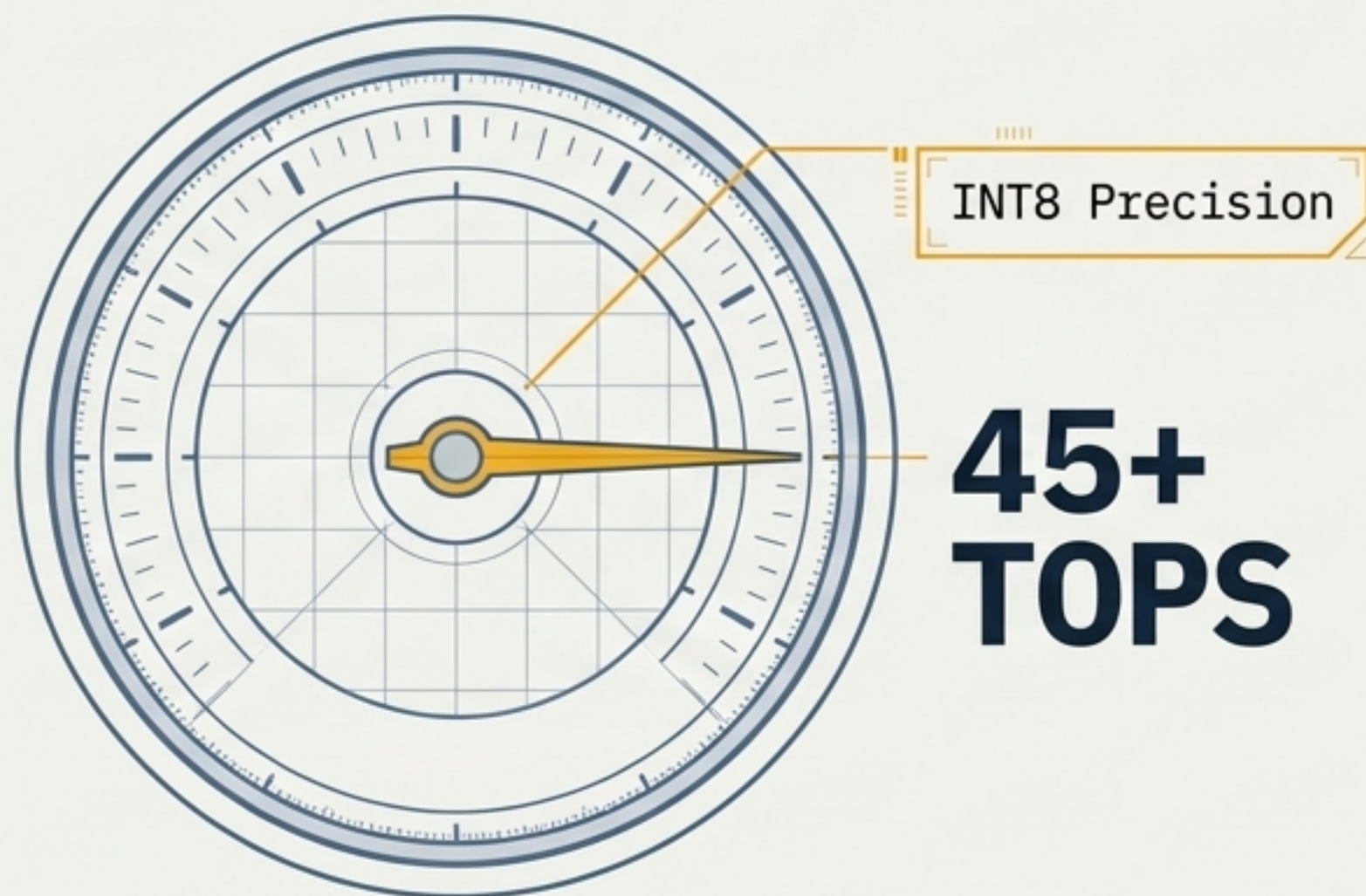
DAM/DMA engines act as intelligent traffic controllers. They quietly and asynchronously prefetch the next “tile” of the AI model from memory while the current tile is being computed.

The Result

Zero downtime. The data flow architecture ensures the assembly line runs around the clock, hiding memory latency and maximizing hardware utilization.

The NPU Advantage: Intelligence at the Edge

The Technical Metric



NPU trade high-precision math (FP32) for low-precision integers (INT8), yielding 10x-100x efficiency gains with virtually no loss in AI accuracy.

The End-User Benefits



Privacy

Data stays on the device. Sensitive biometric or voice data never needs to be sent to a cloud server for processing.



Power

Incredible efficiency. Runs complex models natively on a laptop or smartphone battery, drastically reducing system energy compared to a CPU.



Latency

Real-time, instant processing for video calls, autonomous driving, and proactive voice assistants without network delays.